

Experiment 9

Aim: To apply LIME to find which words in a movie-review comment push a sentiment model toward its positive or negative prediction.

Procedure:

1. Install the libraries. Use distilbert/distilbert-base-uncased-finetuned-sst-2-english model.
2. Download and load the model, and note which label number is NEGATIVE and which is POSITIVE.
3. Write a function that takes a list of sentences and returns the probability of each class. LIME uses this function to talk to the model.
4. Predict the sentence "The plot was weak, but the acting was wonderful." and record the probabilities.
5. Apply LIME and record each word's score.
6. Record every sentence LIME sends to the model. Print the original, 8 random changed copies, and the 3 lowest and 3 highest, each with the model's probability.
7. Remove one word at a time, record the model's probability, and compare with LIME's scores.

Task: Repeat the experiment with a sentence of your own, ideally one containing negation (for example "not good") or a repeated word. Then answer:

1. Which word has the largest positive score, and which the largest negative?
2. Do the removed words in the "lowest" copies match LIME's top words?
3. Does the remove-one-word table rank the words in the same order as LIME?
4. Change the number of samples from 1,000 to 100. Do the scores change?